

Project Design Phase Solution Architecture

Date	24 July 2026
Team ID	Individual / Solo Project
Project Name	Software Installation Request Automation using ServiceNow Service Catalog
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Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Software Installation Request Automation — Architecture Diagram

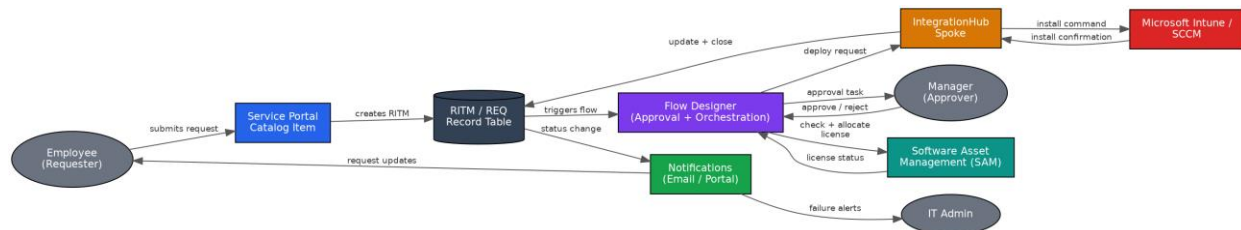


Figure 1: Architecture and data flow of the Software Installation Request Automation system

Component Breakdown

Component	Role
Service Portal Catalog Item	Front-end where employees browse and submit software requests, capturing software name, version, and business justification.
RITM / REQ Table	ServiceNow's native request record structure; stores request details, status, and audit history end-to-end.
Flow Designer	Orchestrates the entire lifecycle: routes approval tasks to managers, checks licensing, and triggers deployment.
Software Asset Management (SAM)	Validates and allocates available software licenses before a request proceeds to fulfillment.
IntegrationHub Spoke	Bridges ServiceNow with external endpoint management tools to trigger real installations.
Microsoft Intune / SCCM	Performs the actual silent software installation on the employee's device and reports back success/failure.
Notifications	Sends email/portal updates to the employee on approval, rejection, and completion; alerts IT on deployment failures.

Data Flow Summary

An employee submits a request through the Service Portal, which creates a RITM record and triggers a Flow Designer flow. The flow routes an approval task to the requester's manager. Once approved, the flow queries SAM to confirm and allocate an available license, then calls an IntegrationHub spoke that instructs Microsoft Intune or SCCM to silently install the software. A confirmation callback updates the RITM, closes the request, and triggers a notification back to the employee — with failure alerts routed to IT if the deployment does not succeed.

Reference: *Adapted from ServiceNow IntegrationHub, Flow Designer, and Software Asset Management documentation patterns.*